

Jongwoo Kim

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Daejeon, Republic of Korea

OBJECTIVE

Coming Soon

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** Aug. 2022 – Present
Ph.D. in Industrial and Systems Engineering; M.S. in Data Science Daejeon, South Korea
 - **Ph.D. in Industrial and Systems Engineering** (Aug. 2024 – Present)
 - * Advisor: Prof. Mun Yong Yi
 - **M.S. in Data Science** (Sep. 2022 – Aug. 2024)
 - * Thesis: *MicroMIL: Spatially Enhanced Multiple Instance Learning for Advanced Cancer Diagnosis in Microscopy*
 - * Advisor: Prof. Mun Yong Yi
- **Seoul National University of Science and Technology** Feb. 2016 – Aug. 2022
B.S. in Industrial and Systems Engineering (Summa Cum Laude) Seoul, South Korea
 - Thesis: *How to succeed in the market? Predicting startup success using a machine learning approach*
 - Advisor: Prof. Youngjung Geum

INTERNATIONAL PUBLICATIONS AND JOURNAL

C=CONFERENCE, J=JOURNAL

† Equal contribution.

- [C.9] Bryan Wong[†], Jongwoo Kim[†], Huazhu Fu, Mun Yi. **Few-Shot Learning from Gigapixel Images via Hierarchical Vision-Language Alignment and Modeling.** In *NeurIPS*, 2025.
- [C.8] Jongwoo Kim[†], Seongyeub Chu[†], Bryan Wong, Mun Yi. **Not All Options Are Created Equal: Textual Option Weighting for Token-Efficient LLM-Based Knowledge Tracing.** In *EMNLP Findings*, 2025.
- [C.7] Jongwoo Kim, Seongyeub Chu, Hyeongmin Park, Bryan Wong, Keejun Han, Mun Yi. **Leveraging Multi-facet Paths for Heterogeneous Graph Representation Learning.** In *CIKM*, 2025.
- [C.6] ChanJae Song[†], Seongyeub Chu[†], Jongwoo Kim[†], Mun Yi. **A Novel Evaluation Framework for 15-Minute City Using Satellite Imagery.** In *ACM SIGSPATIAL*, 2025.
- [C.5] Jongwoo Kim[†], Bryan Wong[†], Huazhu Fu, Willmer Rafell Quiñones Robles, Mun Yi. **MicroMIL: Graph-Based Multiple Instance Learning for Context-Aware Diagnosis with Microscopic Images.** In *MICCAI*, 2025.
- [C.4] Seongyeub Chu[†], Jongwoo Kim[†], Bryan Wong, Mun Yi. **Rationale Behind Essay Scores: Enhancing S-LLM's Multi-Trait Essay Scoring with Rationale Generated by LLMs.** In *NAACL Findings*, 2025.
- [C.3] Seongyeub Chu[†], Jongwoo Kim[†], Mun Yi. **Think Together and Work Better: Combining Humans' and LLMs' Think-Aloud Outcomes for Effective Text Evaluation.** In *CHI, Oral*, 2025.
- [C.2] Hyeongmin Park, Sungrae Hong, Chanjae Song, Jongwoo Kim, Mun Yi. **Uncertainty-based Data-wise Label Smoothing for Calibrating Multiple Instance Learning in Histopathology Image.** In *WACV, Oral*, 2025.
- [C.1] Jongwoo Kim, Minkyu Jeong, Hyeongmin Park, Mun Yi. **Leveraging Spatial Relationships in Microscope for Patient Cancer Diagnosis.** In *ISBI*, 2024.
- [J.1] Jongwoo Kim, Hongil Kim, Youngjung Geum. **How to succeed in the market? Predicting startup success using a machine learning approach.** *Technological Forecasting & Social Change (TFSC)*, 2023. (IF=13.3)

UNDER REVIEW PAPERS

S=IN SUBMISSION / UNDER REVIEW

- [S.6] Jongwoo Kim, Seongyeub Chu, Jeongwhan Choi, Noseong Park, Mun Yi. **GraPE: Graph Prompt Tuning via Mixture of Experts Approach.** Preprint, 2025.
- [S.5] Jeongwhan Choi, Jongwoo Kim, Woosung Kang, Noseong Park. **Learning Posterior Predictive Distributions for Node Classification from Synthetic Graph Priors.** Preprint, 2025.
- [S.4] Jeongwhan Choi, Minseo Kim, Jongwoo Kim, Woosung Kang, Noseong Park. **Can TabPFN Compete GNNs for Node Classification Using Positional and Structural Features?** Preprint, 2025.
- [S.3] Willmer Rafell Quinones Robles, Sakonporn Noree, Bryan Wong, Jongwoo Kim, Mun Yi. **Spatial Context-Driven Positive Pair Sampling for Enhanced Histopathology Image Classification.** Preprint, 2025.
- [S.2] Seongyeub Chu, Jongwoo Kim, Mun Yi. **AMR: Aspect-Aware Mooc Recommendation in a Heterogeneous View.** Preprint, 2025.
- [S.1] Jongwoo Kim, Seongyeub Chu, Mun Yi. **RFMRec: Enhancing Personal Recommendation Using Public Trend Captured via RFM.** Preprint, 2025.

PROJECTS

- **Semiconductor Defect Detection with ETRI & Bluetile Lab (Project Manager)** Sep. 2025 – Present
Tools: Vision Transformer, Weakly Supervised Learning
 - Developed a defect detection pipeline for semiconductor wafer images with **ETRI** and **Bluetile Lab**.
- **Large Vision-Language Model for Pathology Diagnosis with NPF (Project Manager)** Sep. 2025 – Present
Tools: Vision Language Model, Human Computer Interaction
 - Developed a Human-Centered VLM pipeline for histopathology.
- **Personalized Problem Recommendation with Chunjae Education** Feb. 2025 – Sep. 2025
Tools: Knowledge Tracing, Graph Embedding
 - Built a personalized problem recommendation system using **Knowledge Tracing** and **Transition Graph**.
- **Multi-view Essay Scoring with KT Corporation** Jul. 2024 – Dec. 2024
Tools: Transformers, BERTScore
 - Designed a **multi-view essay scoring model** integrating linguistic, semantic, and stylistic features with **KT**.
- **Colon Cancer Diagnosis with Seegene Medical Foundation (Project Manager of Microscope)** Jul. 2023 – Jul. 2024
Tools: Multiple Instance Learning, GNN
 - Developed a **graph-based MIL model** for colon cancer diagnosis using whole-slide microscopy images.
- **Seoul Real Estate Price Prediction with Tushin** Feb. 2023 – Jul. 2023
Tools: P XGBoost, LightGBM, SHAP, Geospatial Analysis
 - Built a **real estate price prediction model** using public datasets and transaction records in collaboration with **Toojae-shin**.

HONORS AND AWARDS

- **Superior Academic Performance** Aug.2020–Fed.2022
Seoul National University of Science and Technology, Korea
- **Capstone Design (2nd Place)** Dec. 2021
Seoul National University of Science and Technology, Korea
- **ETRI Human Understanding Artificial Intelligence Paper Contest (Minister of Science and ICT Award)** Jun. 2022
ETRI, Korea
- **Industrial and Social Issues Poster Contest (2nd Place)** Nov. 2022
KAIST, Korea
- **Best Award of Paper** Jun. 2024
Korea Computer Congress, Korea
- **Industrial and Social Issues Poster Contest (3rd Place)** Nov. 2025
KAIST, Korea

PROFESSIONAL SERVICES

- **Conference Reviewer**
 - ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2024
 - International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2025

REFERENCES

1. **Reference Person 1**
Job Title, Department
Organization/Institution Name
Email: email1@example.com
Phone: +X-XXX-XXX-XXXX
Relationship: [e.g., Thesis Advisor, Manager, etc.]
2. **Reference Person 2**
Job Title, Department
Organization/Institution Name
Email: email2@example.com
Phone: +X-XXX-XXX-XXXX
Relationship: [e.g., Project Supervisor, Colleague, etc.]
3. **Reference Person 3**
Job Title, Department
Organization/Institution Name
Email: email3@example.com
Phone: +X-XXX-XXX-XXXX
Relationship: [e.g., Mentor, Collaborator, etc.]